4/13/2026

SAUDI ARABIA EMPLOYEE ATTRITION DATASET

MINI PROJECT

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1.PROJECT OVERVIEW AND OBJECTIVES

This project focuses on analyzing employee attrition using a structured HR dataset collected from organizations in Saudi Arabia. The dataset includes employee demographic details, job-related information, compensation factors, training participation and psychological aspects such as job satisfaction, stress, and emotional commitment.

The raw dataset contains mixed formats, categorical values, and inconsistent entries. Data cleaning done in Excel via Power Query Editor to remove duplicates, standardize text values, handle missing or unclear information. Also categorical data converted into numerical data like Attrition Flag, Salary Level and Useful Training Flag to improve the analysis.

After cleaning the data loaded into Power BI to build multiple interactive dashboards covering demographics, compensation, training, and employee satisfaction. These dashboards help to identify both individual and combined effects of variables, key factors influencing attrition and supports the data driven HR decisions.

1.Analyze the impact of demographic characteristics like Gender, Marital Status and Age Category on Attrition

2.Evaluate how commuting distance influence Attrition

3. Examine the impact of job-related factors through Job Title, Department and Experience.

4. Assess how the Compensation factors (Salary, Bonus) and Workload (Overtime) affects Attrition.

5. Analyze the Influence of Training and Development

6. Identify Satisfaction and Recognition segments such as Rewards-wages and Promotion.

7. Analyze the Work Life Balance measures through Physical stress and job engagement.

2. DATA SOURCE

Source Description and Timeline:

https://data.mendeley.com/datasets/6z2hty8php/1/files/b422d0af-74b6-47b8-94ff-614e09c2382f

and 19.03.2025.

Domain: Human Resource Analytics

3.PROBLEM STATEMENT

Employee attrition is a major challenge for organizations, leading to increased recruitment costs, loss of skilled talent, and reduced productivity. This project aims to analyze employee data from various sectors in Saudi Arabia to identify the key factors influencing employee attrition.

There is no clear visibility into which factors contribute most to employees leaving the organization, such as job satisfaction, workload and job roles. The lack of interactive analysis makes it difficult to compare attrition across departments, roles, and demographics.

These factors are not directly structured for analysis and require cleaning, transformation, and feature engineering. The problem is to determine how variables such as salary, job role, department, training, stress levels, and job satisfaction contribute to employee attrition and to provide actionable insights for improving employee retention strategies.

4. **ATTRIBUTE DETAILS**

Attribute Name	**Data Type**	**Description**
Employee_ID	Integer	Unique ID assigned to each employee
Attrition	Text (Yes/No)	Indicates whether employee left the company
Gender	Text	Gender of employee
Age	Text (Categorical)	Age group of employee
Marital_Status	Text	Marital status (Single/Married)
Academic_Degree	Text	Highest education qualification
Years_Experience	Text	Total years of work experience
Sector	Text	Industry sector of employee
Department	Text	Department where employee works
Job_Title	Text	Employee job role
Monthly_Salary	Text	Salary range of employee
Allowances	Integer	Count of Additional allowance amount
Bonus	Text (Yes/No)	Whether employee receives bonus
Over Time	Text (Yes/No)	Overtime work status
Rewards_Wages_Satisfaction	Text (Yes/No)	Satisfaction with salary and rewards
Get_Deserved_Promotion	Text (Yes/No)	Whether employee feels promotion is deserved
Training_Programs	Text	Number/range of training programs attended
Useful_Training_Programs	Text (Yes/No)	Whether training was useful
Job_Support	Text	Level of support at workplace
Emotional_Commitment	Text	Emotional attachment to job
Job_Engagement	Text	Level of engagement at work
Distance_To_Work	Text	Distance from home to workplace
Work_Life_Balance	Text	Work-life balance level
Physical_Stress	Text	Level of physical stress

5. **TOOLS & TECHNOLOGIES**

Microsoft Excel:

- Data Cleaning
- Performed logical functions and Statistical functions
- Applied conditional formatting

Power Query:

- Removing Duplicates
- Replaced null values
- Datatype Formatting
- Created calculated columns like Age category, Salary Numeric
- Renamed and Merged the columns

Power BI:

- Data Modeling
- DAX calculations
- Visualizations
- Interactive dashboard creation

6. DATA PRE-PROCESSING (EXCEL / POWER QUERY)

Tasks Performed in Excel: Logical functions like IF () and Nested IF () is used to calculate the numerical columns for

- Age
- Attrition,
- Marital_Status
- Distance_To_Work
- Seniority

Conditional Formatting applied for the Attrition column

Statistical functions were used to calculate total count, Attrition count, Attrition rate, Attrition based on many categories.

Tasks Performed in Power Query:

- Duplicates on Employee_ Id (Primary key) are removed
- Replaced the inconsistent values and Standardized the text format
- Merged Sector and Department column into one (Sector-Department)
- Datatype formatting done for each column

- Created calculated columns for Salary, Experience and Age Category since they were categorical

Data Modeling:

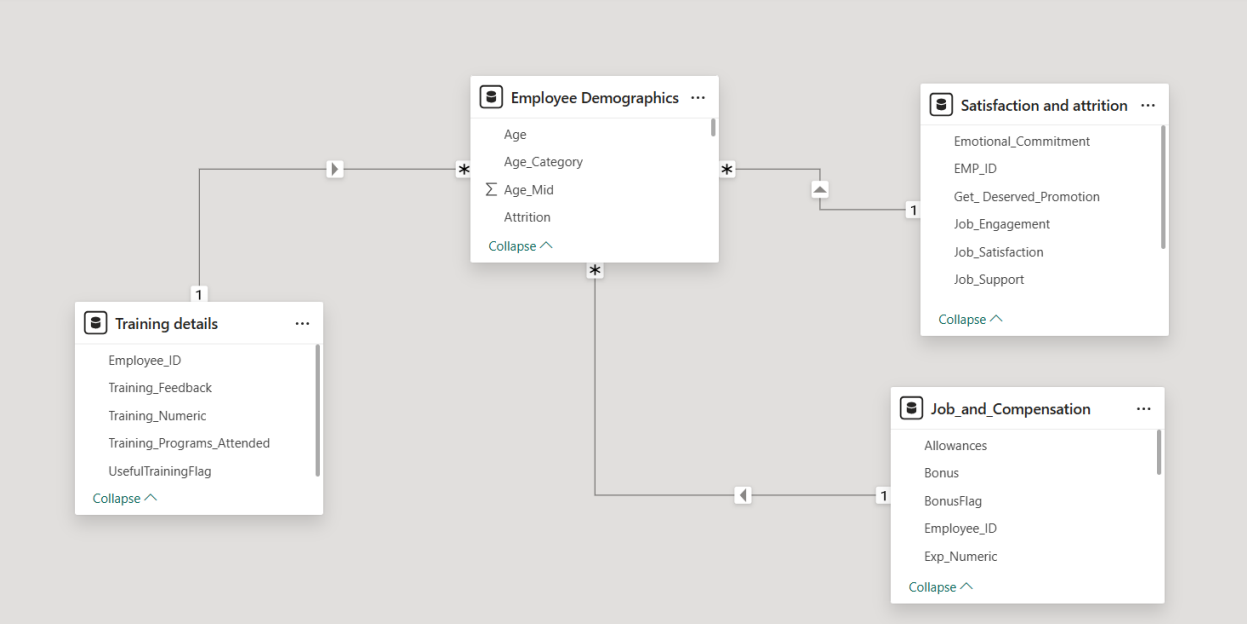
The raw dataset was cleaned and structured for analysis and were broken down into four normalized tables

Named

- Employee (Fact Table)
- Job and Compensation (Dimension Table)
- Training Details (Dimension Table)
- Satisfaction and Attrition (Dimension Table)

to improve data organization, reduce redundancy and enable efficient analysis.

7. DATA MODELLING AND DAX (POWER BI)



The data model is designed using a Star Schema, with the Employee table as the central fact table. It is connected to multiple dimension tables such as job and compensation, training details and satisfaction & attrition for analysis.

Established

One-to-Many (1: *) relationships with single direction filtering.

Calculated Columns:

Created calculated columns for Satisfaction Score and Overtime Category for better analysis.

DAX Measures:

Total Employees - total number of employees in the dataset

Attrition Count - total number of employees who have left the organization

Attrition Rate (%) – Percentage of employees who left the organization

Active Employees – Employees currently working in the organization

Avg Salary - Average Salary of all employees

Overtime Employees - number of employees who work overtime

Overtime (%) - percentage of employees who work overtime

Bonus (%) - percentage of employees who received a bonus

Useful Training Count - Counts the number of employees who said training was useful.

Useful Training % - percentage of employees who found their training useful out of total employees.

Not Useful Training Count - Counts the number of employees who said training was not useful.

Not Useful Training (%) - percentage of employees who found training not useful out of total employees.

Satisfied Employees - number of employees who are satisfied with their job.

Satisfaction % - Percentag**e** of employees who are satisfied with their job out of total employees.

High Stress Employees - number of employees experiencing high physical stress.

Stress % - Percentage of employees experiencing physical stress out of total employees.

8.ANALYSIS AND VISUALIZATIONS (Power BI)

Demographics Analysis:

Gender- Attrition is more among Female compared to male.
Age Category- Junior and Mid -level employees represent the higher attrition groups.
Marital Status- Married employees have the highest attrition rate among others.
Distance to work - Employees with medium and close distances exhibit higher attrition rate.
Age- Employees around the age of 25 represent highest attrition rate compared to other age groups.

Compensation Analysis:

Job Title- Customer Services Representative under various sectors shows high attrition
Sector-Department- Education-Teaching sector exhibits higher attrition
Salary- Low salary employees records high attrition
Experience- Entry level employees are more likely to leave the organization.
Overtime- Employees with no overtime shows high attrition
Bonus- Employees who didn't receive bonus are more likely to leave

Training Analysis:

Trainings attended- Employees who attended less trainings left the organization
Useful Training- Employees who said training useful stayed more compared to attrited employees.

Satisfaction Analysis:

Satisfaction- Satisfied employees shows high attrition compared to very satisfied
Rewards & wages- Employees who didn't get rewards mostly leave the organization
Job Engagement- Medium level engagement show high attrition
Promotion- Employees who didn't get promotion exhibits high attrition
Physical stress- Employees who experience high physical stress more likely leave

Visualizations:

1.Multi-Page Navigation (Bookmarks-buttons)

Dashboard contains four pages

- Demographics
- Compensation
- Training
- Satisfaction

Navigation is implemented using Bookmarks + Buttons

2.Sync Filtering (Slicers)

Slicers are synchronized across all report pages to ensure consistent filtering of data throughout the dashboard.

Filters available:

- Job-Title
- Sector Department

3.**Bookmarks Creation**

A bookmark was created to display male employees with Attrition = Yes by applying filters and saving the report state using Power BI bookmark functionality with Data enabled.

4.KPI Cards (Key Metrics)

Displays key metrics like

- Total Employees
- Attrition count
- Attrition rate
- Average salary
- Overtime %
- Satisfaction %
- Stress %

5.**Clustered Bar Chart**

Shows the attrition count based on Gender (Male / Female)

6. **Stacked Column Chart**

Displays the attrition count by several categories like

- Age Category
- Salary
- Trainings attended
- Promotion

7.**Line Chart**

It is used to identify attrition trends and patterns across

- Age
- Experience

8. Pie Chart

Pie chart is used to show the proportion of employee attrition across different categories like

- Marital Status
- Overtime
- Stress

9. **Stacked Bar Chart**

A stacked bar chart is used to show attrition count across several categories like

- Distance_To_Work

- Job Title
- Bonus
- Rewards & Wages

10. **Donut Chart**

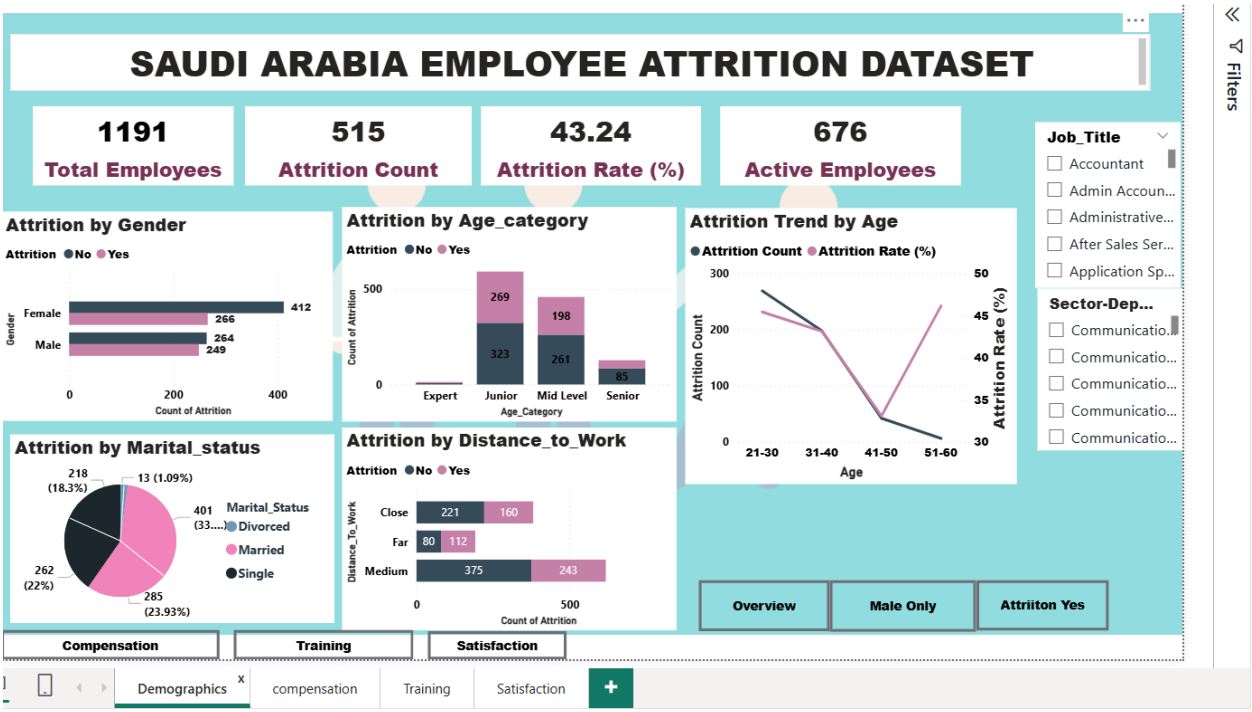
A donut chart is used to show the distribution of employees with attrition count and attrition status by

- Useful Training flag
- Satisfaction score

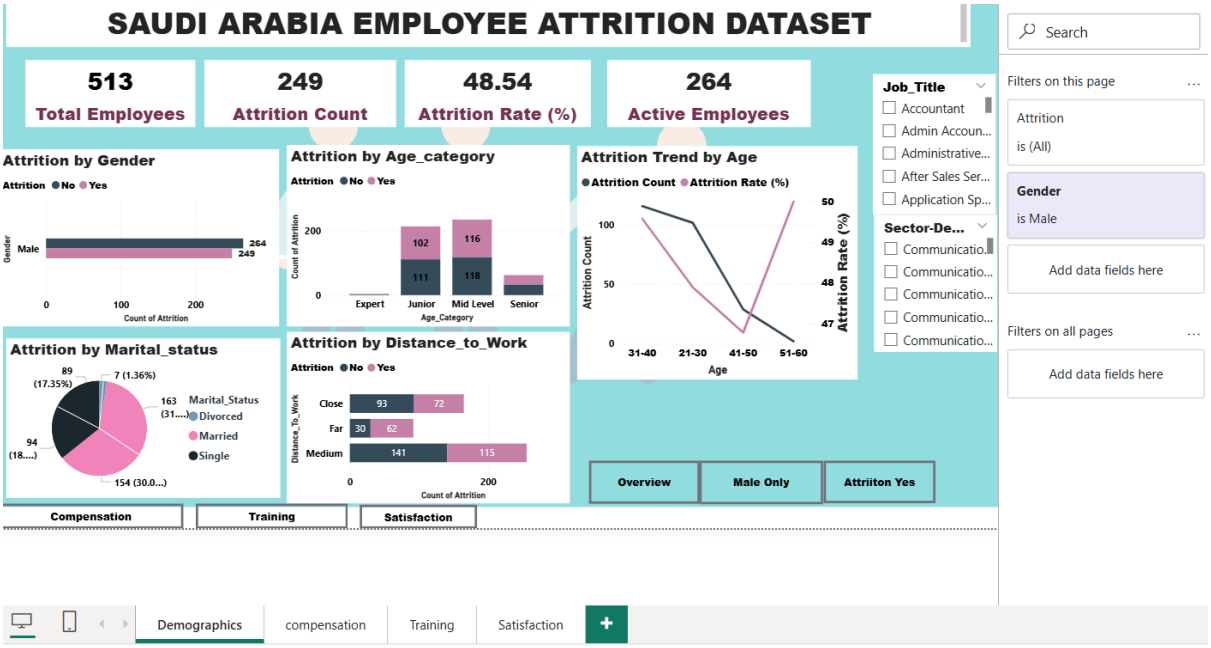
11.**Treemap**

A Tree map is used to visualize the **attrition count across different departments**, where each department is represented as a rectangle and the size of the rectangle indicates the number of employees who have left the organization.

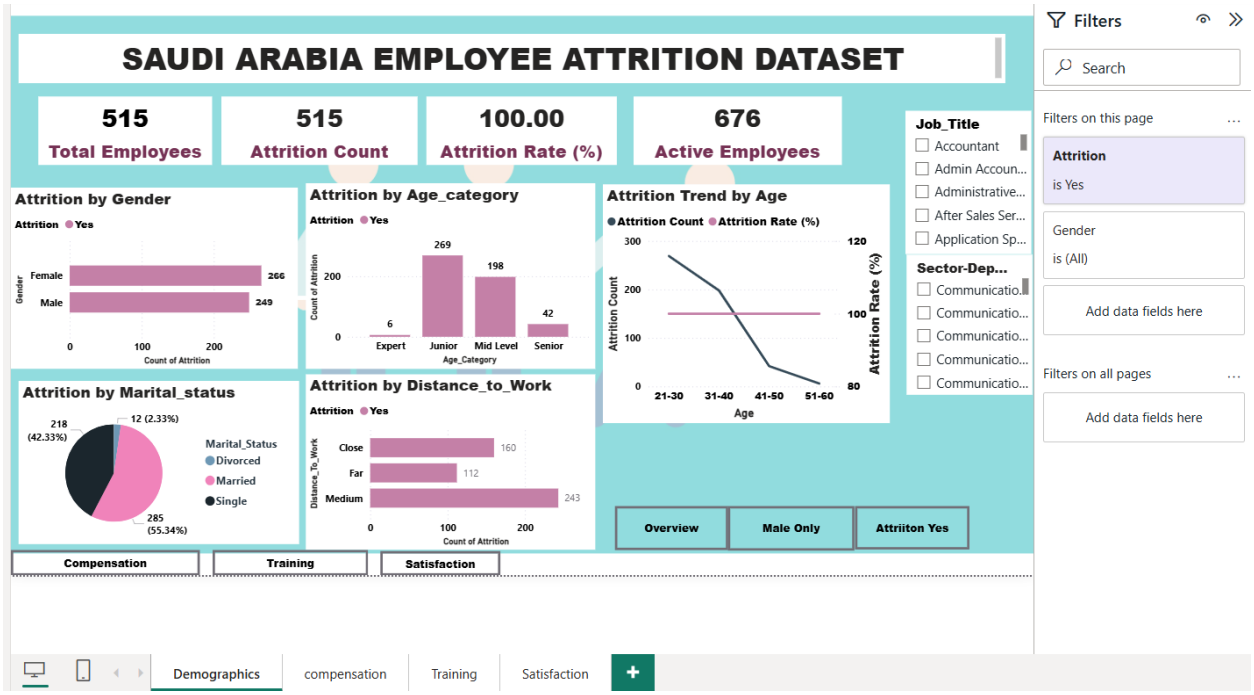
REPORT SCREENSHOTS:



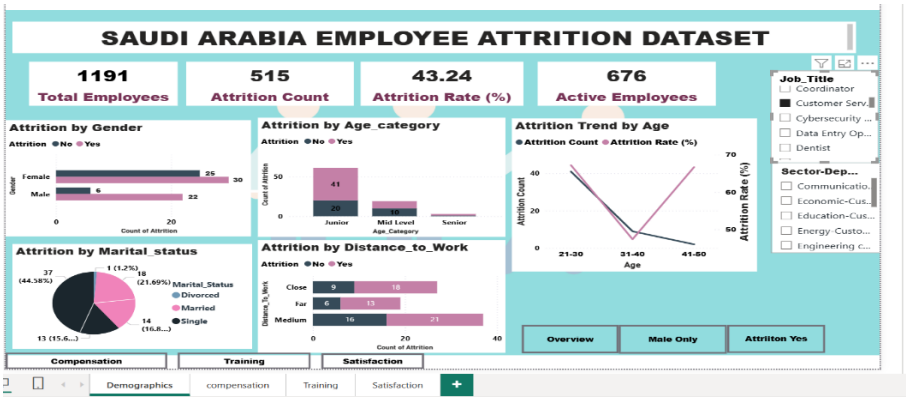
First page of report- Demographics analysis



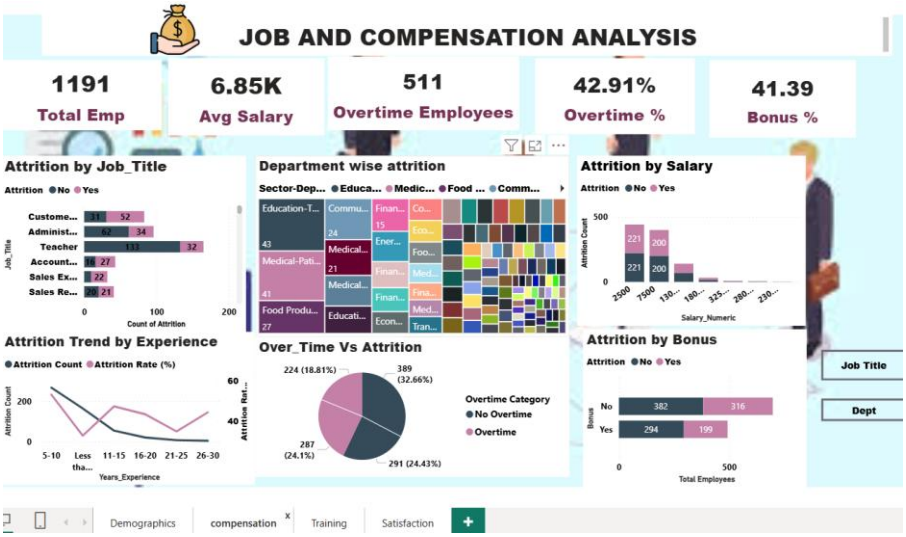
Bookmark affecting Male only in Gender



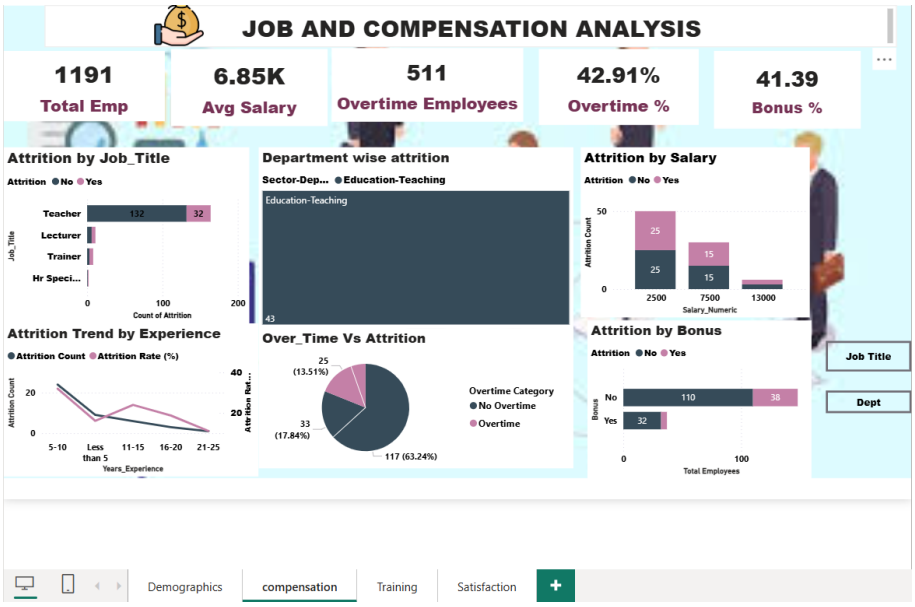
Bookmark affecting Attrition “Yes”.



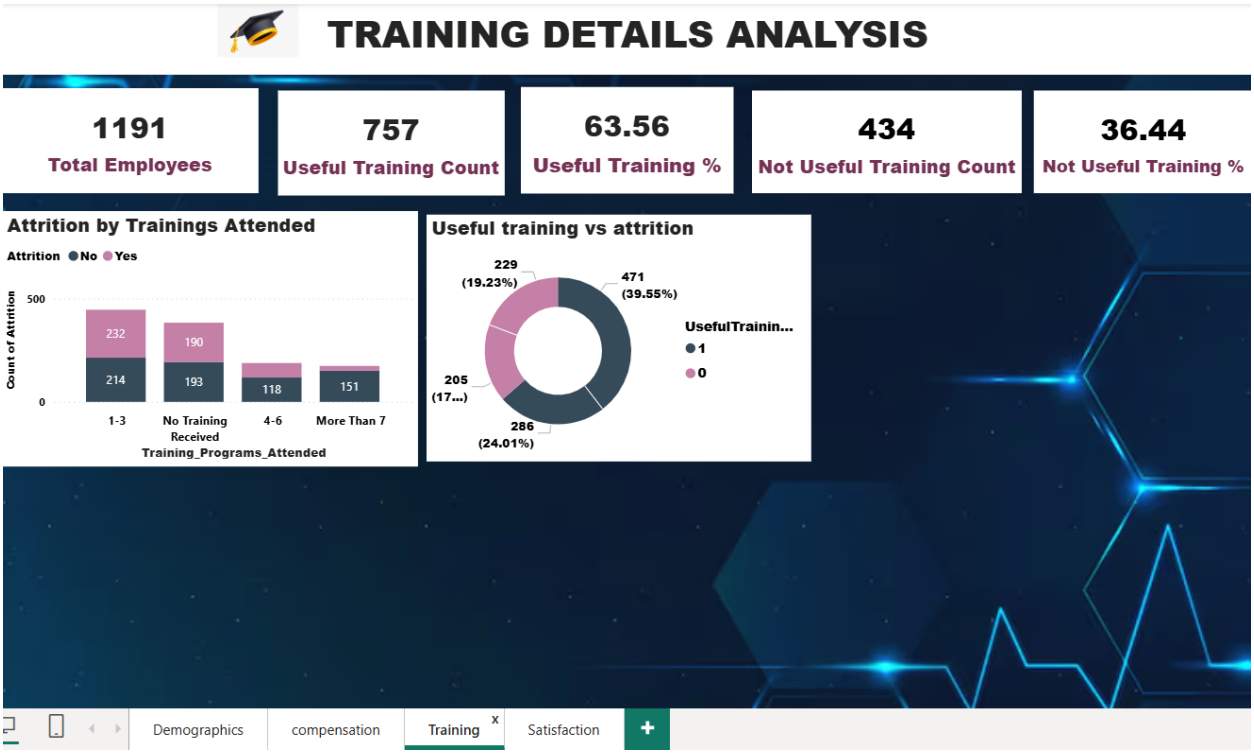
Sync-Filters for Job Title in first page of report affecting for Customer Services Representative



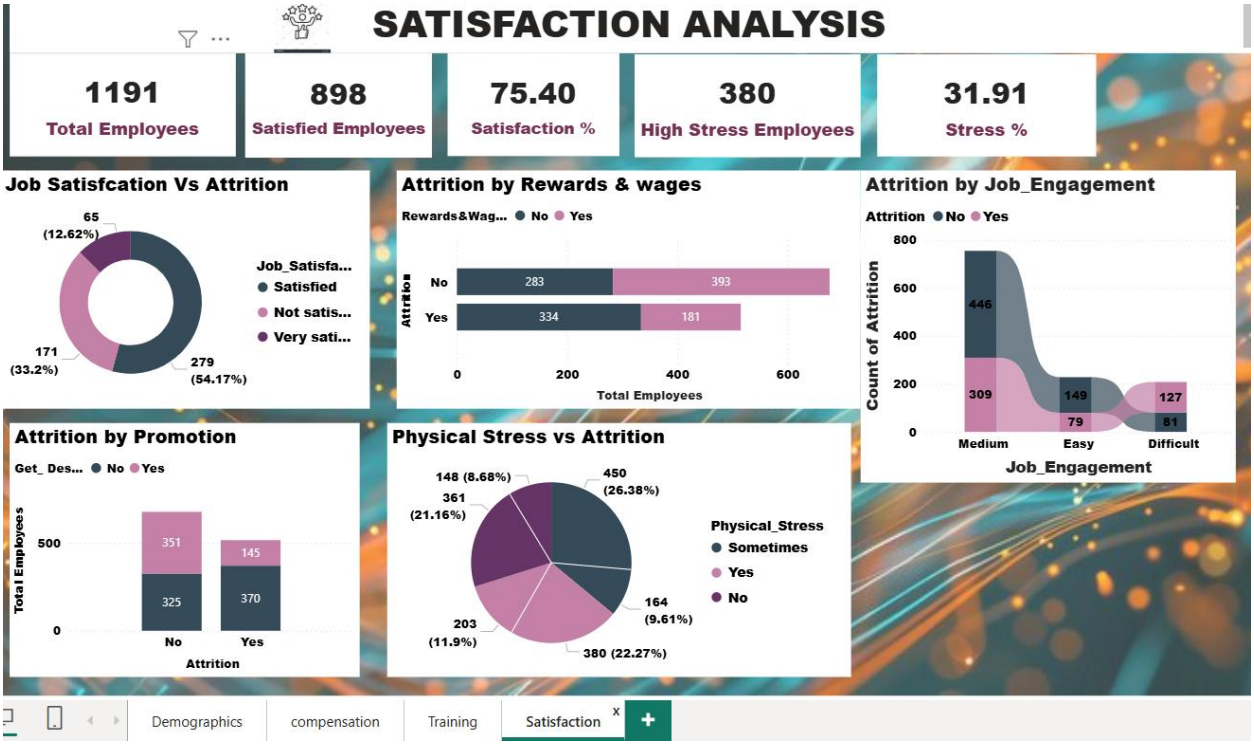
Second page -Compensation analysis



Second page affecting for sync filter in first page for sector -department (Education-Teaching)



Third page – Training analysis



Fourth page- Satisfaction analysis

9. **INSIGHTS**

Demographics analysis:

S.NO	CATEGORY	KEY FINDINGS ON ATTRITION	INSIGHTS
1.	Gender	Female employees (266) Male employees (249)	indicate work-life balance concern
2.	Age Group	Junior (269) Mid-level-198	Entry level employees leave more
3.	Marital Status	Married (285) Single (218)	Family responsibilities may influence
4.	Age	(21-30) – 269 (31-40) - 198	leaving due to limited growth opportunities
5.	Distance to work	Medium – 243 Close – 160	Medium distance employees are affected more than close

Compensation analysis:

S.NO	CATEGORY	KEY FINDINGS ON ATTRITION	INSIGHTS
1.	Job Title	Customer services representative (52)	May indicate limited career growth
2.	Sector-Dept	Education-Teaching (43)	Shows workload pressure
3.	Salary	Less than 5000- 221 5000-10000 - 200	Shows financial dissatisfaction
4.	Experience	Less than 5 years – 266 (5-10) years - 164	Indicates more job switching opportunities
5.	Overtime	Yes – 224 No - 291	Lack of involvement in work
6.	Bonus	Yes – 199 No - 316	Dissatisfaction with rewards and recognition

Training Analysis:

S.NO	CATEGORY	KEY FINDINGS ON ATTRITION	INSIGHTS
1.	Attended trainings	(1-3)– 232 (No training) – 190 (4-6) – 70 More than 7 - 23	More trainings will reduce the attrition count It improves the engagement
2.	Trainings useful or not	Useful but not attrited (471) Useful but attrited – (286)	Not Attrited count is more than attrited one Indicates training is effective

Satisfaction Analysis:

S.NO	CATEGORY	KEY FINDINGS ON ATTRITION	INSIGHTS
1.	Satisfaction	Satisfied- 279 Very satisfied – 65 Not satisfied – 171	Satisfied is more than others. Shows some other risks but not satisfied still shows high attrition count
2.	Rewards and wages	Yes- 181 No - 334	Mostly shows no rewards high attrition
3.	Job Engagement	Medium – 309 Easy – 79 Difficult – 127	It shows not fully committed to the work
4.	Promotion	No – 370 Yes - 351	Reduces career progression motivation
5.	Physical Stress	Yes- 203 Sometimes – 164 No - 148	High stress indicates more attrition

Descriptive: Attrition is higher among young, junior, low-salary, and medium-distance employees with low promotion, bonus, and engagement.

Diagnostic: Employees leave mainly due to low pay, limited career growth, workload stress, and lack of engagement.

Predictive: Early-career, low-paid, and less-engaged employees are likely to continue showing higher attrition.

Prescriptive: Improve salary, promotions, rewards, engagement, training, and workload balance to reduce attrition.

10.CONCLUSION:

Employee attrition is mainly driven by young and junior employees, low salary, and limited career growth opportunities. Lack of promotion, low rewards, and moderate job engagement also increase employee turnover. Training and satisfaction levels play a key role in improving employee retention.

Work-related factors like stress and commute distance further influence attrition decisions. Organizations need to focus on better compensation, engagement, and career development. Overall, improving employee experience can significantly reduce attrition rates.